

∇p_0 H_0 : Все ученики равнообразны $(p_i = \frac{1}{10})$
 $a)$ $H_1: \bar{H}_0$

m_i : 5 8 6 12 14 18 11 6 13 7

np_i : 10 10

$$n = 100$$

$$p_i = \frac{1}{10}$$

$$\tilde{\Delta} = \sum_{i=0}^9 \frac{(m_i - np_i)^2}{np_i} \approx 16,4$$

$$\Delta \rightsquigarrow \chi^2(10 - 1)$$

$$\Delta \rightsquigarrow \chi^2(9)$$

$$p\text{-value} = P(\Delta \geq \tilde{\Delta} | H_0) = \int_{16,4}^{+\infty} q(t) dt \approx 0,059$$

$$p\text{-value} > \alpha$$

$\Rightarrow$ Нет оснований отвергнуть H_0 .

б) ОДПГ.

$$H_0: \xi \sim N(\theta_1, \theta_2)$$

$$H_1: \bar{H}_0$$

$A_i (-\infty; 1); \dots \dots \dots [0; +\infty)$

m_i 5 ; 8 ; 6 ; ~~12~~ ; 14 ; 18 ; 11 ; 6 ; 13 ; 7

hp_i 5, 47 ; 5, 51 ; 8, 66 ; 11, 87 ; 14, 18 ; 14, 76 ; 13, 38 ; 10, 58 ; 7, 28 ; 8, 31

(Правильно есть в коде)

$$p_i(\theta_1, \theta_2) = \int_{A_i} \frac{1}{\sqrt{2\pi}\theta_2} \exp\left[-\frac{(x-\theta_1)^2}{2\theta_2^2}\right] dx$$

$$L(\theta_1, \theta_2) = \prod_{i=0}^g p_i(\theta_1, \theta_2)^{m_i}$$

$$\ln L(\theta_1, \theta_2) = \sum_{i=0}^g m_i \ln(p_i(\theta_1, \theta_2))$$

→ max
 θ_1, θ_2

$$\tilde{\theta}_1 \approx 5,2897$$

$$\tilde{\theta}_2 \approx 2,6795$$

$$\tilde{\sigma} = \sum_{i=0}^g \frac{(m_i - np_i)^2}{np_i} \approx 9,8026$$

$$\Delta \sim \chi^2(7)$$

$$p\text{-value} = P(\Delta \geq \tilde{\sigma} | H_0) \stackrel{10}{=} \stackrel{1-2=7}{=} \int_{9,8026}^{\infty} q(t) dt \approx 0,20004$$

$$p\text{-value} \approx 0,20004 > 0,2$$

⇒ keine Ablehnung H_0 .